



Regional forecasting of significant wave height and mean wave period using EOF-EEMD-SCINet hybrid model

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ABSTRACT

Most studies only focus on single-point wave forecasting, while regional wave forecasting has more important significance for ocean engineering construction, navigation safety and disaster warning. To solve the problems of high computational cost and poor performance of regional wave forecasting models, a novel hybrid model for regional significant wave height (SWH) and mean wave period (MWP) forecasting, called EOF-EEMD-SCINet, is proposed by combining empirical orthogonal functions (EOF) analysis, ensemble empirical mode decomposition (EEMD) and sample convolution and interaction network (SCINet). After determining the optimal input steps, the number of intrinsic mode functions (IMFs) and the decomposition method, the forecasting performance of the model for SWH and MWP with lead times of 24, 48 and 72 h is evaluated based on SWH, MWP, wind speed (WSPD), significant height of first swell partition (SWH1) and significant height of wind waves (SHWW) in the South China Sea (SCS). The results show that the proposed model can not only accurately forecast the changes in SWH and MWP with time, but also has a high forecasting precision for regional waves, which is superior to other models. In addition, the increase of wave lead time has less negative effect on the forecasting performance of the proposed model compared to other models. As the lead time changes from 24 h to 72 h, the mean absolute error (MAE) of SWH increases by less than 6 cm, and the MAE of MWP changes around 0.04 s. Besides, EOF-EEMD-SCINet has enormous advantage in terms of efficiency. EOF-EEMD-SCINet is a model that can simultaneously forecast regional SWH and MWP with high precision.

1. Introduction

As important wave parameters, significant wave height (SWH) and mean wave period (MWP), their precise forecasting is not only necessary for ocean engineering construction, navigation safety and disaster warning (Jain and Deo, 2006; James et al., 2018), but also very important for the development and utilization of wave energy (Cuadra et al., 2016). However, the complexity of ocean conditions and the strong nonlinearity of waves pose great challenges to the accurate prediction of SWH and MWP. In general, wave forecasting methods are mainly divided into four categories (Yang et al., 2021): numerical models, classical time series models, machine learning models and hybrid models.

Currently, numerical wave models based on the energy balance equation are widely used for SWH and MWP forecasts. The mainstream numerical models include the third-generation wave model (WAM) (Group, 1988), simulating waves nearshore model (SWAN) (Booij et al.,

1999) and WAVEWATCH III (WW3) (Tolman et al., 2002) based on WAM. However, due to the high complexity of input and calculation, numerical models require not only expensive data and high-performance computers, but also a lot of computational time. Therefore, the application of numerical models in wave forecasting is limited (Chang et al., 2011). Considering the shortcomings of numerical models, time series models such as autoregressive model (AR), autoregressive moving average model (ARMA), and autoregressive integrated moving average model (ARIMA) are often used for wave forecasting. Fusco (2009) used AR to forecast SWH, and the results showed that AR was suitable for predicting the waves of more than one wave period. Time-varying regression was proposed by Reikard (2009) to be applied to the forecasting of wave height and period, but the prediction effect was poor. Moreover, Agrawal and Deo (2002) compared the prediction performance of ARMA and ARIMA in an experiment, which showed that the results of the two forecasting models were not very different. However, time series models have the shortcomings of poor forecasting

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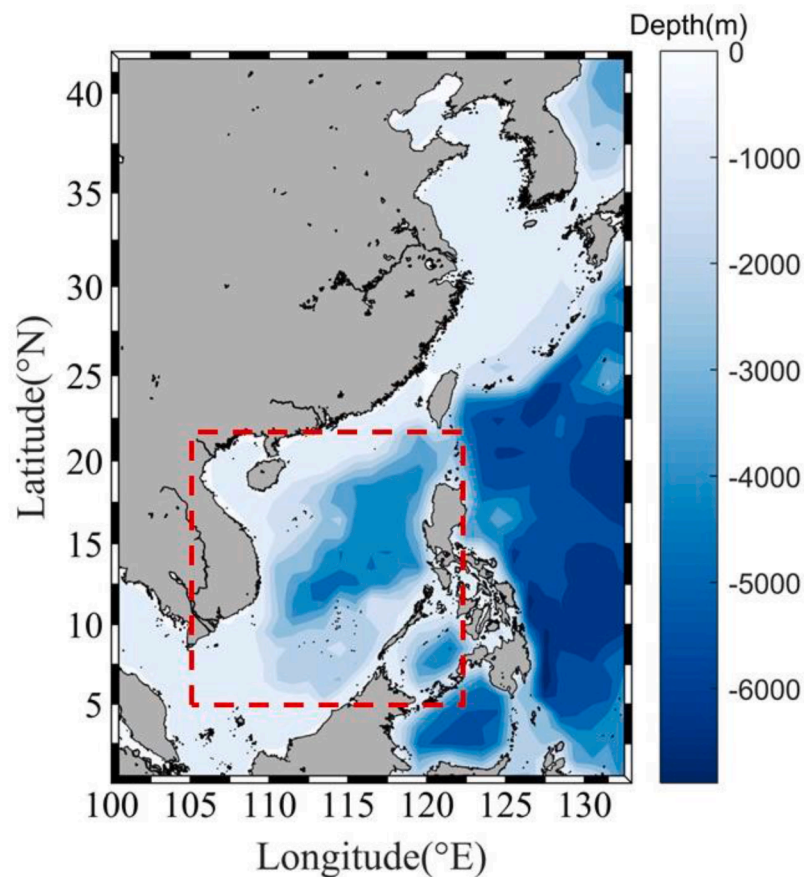


Fig. 1. The study area and bathymetry. The red dashed rectangular box represents the study area. The bathymetric data are sourced from ETOPO1.

performance for longer lead times and necessary stationary hypothesis.

Machine learning models have the advantages of low computational cost, high precision and high efficiency, so there are applications of machine learning in various fields (Campos et al., 2020; Minuzzi and Farina, 2023; Zhou et al., 2022). In recent years, many researchers have used machine learning in wave forecasting studies and achieved good results. Malekmohamadi et al. (2011) studied the performance of several models to forecast SWH, including support vector machine (SVM), bayesian network (BN), artificial neural network (ANN), and adaptive neuro-fuzzy inference system (ANFIS). Fan et al. (2020b) applied long short-term memory (LSTM) to SWH forecast, and the results showed that LSTM can not only perform short-term predictions, but also has a strong long-term forecasting ability. Hu et al. (2021) compared the performance of LSTM and extreme gradient boosting (XGBoost) in predicting wave height and period, and they found that the overall performance of XGBoost was better than that of LSTM. Choi et al. (2020) used the original ocean image as input and used convolutional long short-term memory (ConvLSTM) to forecast SWH in real time. To forecast SWH and peak wave period, a physics-based machine learning model was proposed by Wu et al. (2020), but the forecasting precision of the peak wave period was low.

The strong nonlinearity and nonstationarity of wave data often affect the forecasting performance of single models. To solve this problem, hybrid models have attracted the attention of many scholars. Combinations of several models are used for wave forecast to improve forecasting precision. Jain and Deo (2007) first used spatial mapping and the statistical technique of multiple regressions to fill in the missing information, and then used ANN to forecast SWH. Meng et al. (2022) used an adaptive time-frequency decomposition method and a deep learning method to forecast tropical cyclone wave height in real time and achieved good results. Before using LSTM for SWH forecast, Huang and

Dong (2021) used an improved complete ensemble empirical mode decomposition algorithm and recurrence quantification analysis to separate the original data into deterministic and stochastic components to improve the forecasting performance of the model. Seasonal-trend decomposition based on loess (STL), one-dimensional convolutional neural network (CNN) and positional encoding (PE) were combined to forecast multi-hour SWH by Yang et al. (2021). All the results show that hybrid models combined with decomposition algorithms often outperform single models.

Most studies only focus on single-point wave forecasting, while regional wave forecasting is of more importance. Precise forecasting of regional waves helps to master the overall wave situation of a certain region, which is helpful to develop ocean energy and prevent ocean disasters. Compared with single-point wave forecasting, the number of machine learning models and hybrid models applied to regional wave forecasting is limited. Considering the expensive computational cost of the SWAN model, Jing et al. (2022) proposed a CNN-based regional wave prediction (CNN-RWP) model to predict large-scale waves. However, this study only forecasted SWH with lead time of one hour and did not validate the forecasting performance of the model for multi-hour SWH. To forecast the monthly and annual average wave fields of the South China Sea (SCS), Bai et al. (2022) proposed a two-dimensional deep learning regional wave field forecast model based on CNN and demonstrated the long-term forecasting ability of the model, but this model only forecasted one wave parameter, SWH. Shao et al. (2021) proposed a hybrid model combining empirical orthogonal functions (EOF) analysis, decomposition algorithm and ANN for medium-term forecast of multiple variables such as sea surface height, temperature, salinity, and velocity in the SCS, but it focused on forecasting the daily evolution of ocean surface variables.

In this study, a regional wave forecasting model combining EOF

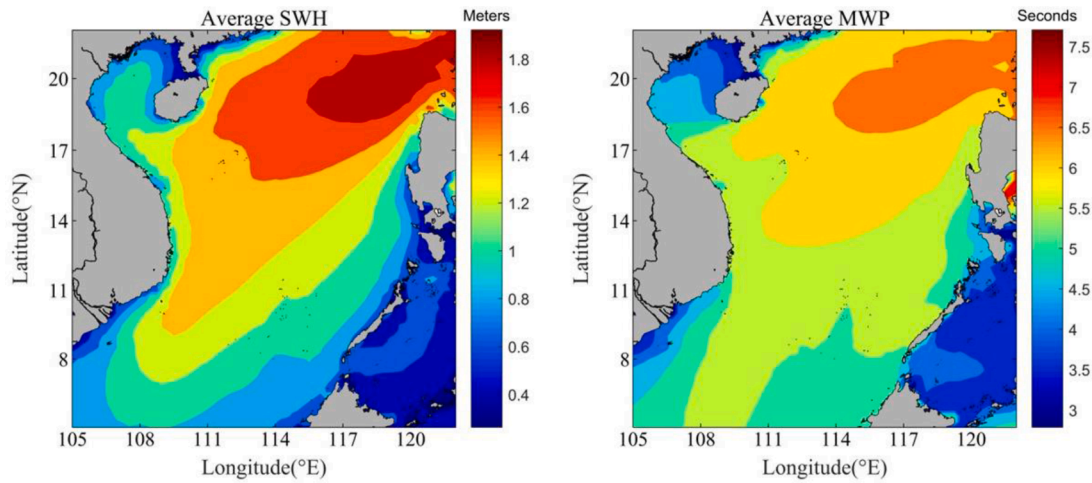


Fig. 2. Average SWH (left) and MWP (right) of the study area in 2020. The data used are provided by ECMWF.

analysis, ensemble empirical mode decomposition (EEMD) and sample convolution and interaction network (SCINet) is proposed and applied to the regional forecast of SWH and MWP in the SCS. SCINet is a novel neural network architecture proposed by Liu et al. (2021) for the time series forecasting problem, and the advantages of this model have been demonstrated using multiple datasets. The hierarchical framework of SCINet can obtain a large number of effective features to enhance the predictability of the series. The essence of using EOF analysis to transform reanalysis data into spatial modes (EOFs) and principal components (PCs) is to transform the spatiotemporal series forecasting problem into a time series forecasting problem. EOF analysis not only reduces the dimensionality of data, but also takes into account the correlation between multiple variables and retains most of the information of the original data. After that, the PCs can be forecasted using a time series forecasting model. To extract more features and improve the forecasting precision, the PCs are decomposed using EEMD. Finally, the forecasts of multi-hour regional SWH and MWP are implemented using SCINet.

The remainder of this paper is structured as follows. Section 2 describes the study area and dataset. Section 3 presents the basic principles of EOF, EEMD and SCINet and the framework of the proposed model. The optimization results of the proposed model are given in Section 4, including the determination of the input steps, the number of intrinsic mode functions (IMFs) and the decomposition method used. In Section 5, the performance of the proposed model in multi-hour SWH and MWP forecasts is assessed. Section 6 elaborates on the conclusions based on the experiments.

2. Study area and dataset

2.1. Study area

As shown in the red dashed rectangular box of Fig. 1, the study area is a sea area (5°N–22°N, 105°E–122°E) in the SCS. The SCS, the largest deep marginal sea in Southeast Asian waters (Wang et al., 2018a), is almost surrounded by continents and islands. The SCS is connected to the Pacific Ocean in the east and the Indian Ocean in the west through waterways or straits. It is a northeast-southwest semi-enclosed sea. Cold air in winter and typhoons in summer result in a unique wave distribution characteristic in the SCS (Wang et al., 2018b). The average depth of the SCS is more than 1000 m, and the maximum depth is more than 5000 m.

2.2. Dataset

Buoy data, satellite altimeter data and reanalysis data are the three most commonly used data for wave parameter analysis. The precision of

buoy data is higher and the quality is better. However, buoy data are affected by the density of buoy station distribution and the number of buoys is limited. Satellite altimeter data have high precision and wide coverage, but their sampling time and space are irregular. Although the precision of reanalysis data is slightly lower, reanalysis data have the advantages of long series, comprehensive consideration of factors and more choice of spatial and temporal resolutions. Therefore, the fifth generation of global climate and weather reanalysis dataset (ERA5) provided by the European Center for Medium-Range Weather Forecasts (ECMWF) is selected as the dataset used in this study. In order to make up the flaw of the uneven spatial and temporal distribution of the observations and the lack of information, ERA5 assimilates all kinds of observations using the four-dimensional variational (4D-VAR) analysis systems. The ERA5 dataset has been widely used in recent years for atmospheric and oceanic studies (Sreelakshmi and Bhaskaran, 2020; Stopa et al., 2013).

To predict SWH and MWP simultaneously, the reanalysis data of multiple parameters in ERA5 are used in this paper, including SWH, MWP, wind speed (WSPD), significant height of first swell partition (SWH1), and significant height of wind waves (SHWW). The time range of the dataset is from January 1, 2020 to December 31, 2020, and the time resolution is 1 h. After bilinear interpolation, the spatial resolution is $0.25^\circ \times 0.25^\circ$. Nine months of data from January 2020 to September 2020 are used for training the model, and three months of data from October 2020 to December 2020 are used for testing the model. The annual average SWH and MWP of the study area in 2020 are shown in Fig. 2. The distribution characteristics of SWH and MWP are roughly the same. Wang et al. (2021) evaluated the precision of SWH reanalysis data in ERA5 using buoy observations of nine stations in the SCS throughout 2020. Each of the nine buoy stations corresponds to the closest grid point. The correlation coefficient (R) is 0.95 and the root mean squared error (RMSE) is 0.35 m. The comparison results show that the metrics are all within reasonable limits. Considering the bilinear interpolation of the reanalysis data and the partial lack of buoy observations, the precision of the ERA5 data is sufficiently high.

3. Methodology

3.1. EOF analysis

EOF analysis, also known as principal component analysis (PCA), is a method for analyzing the structural features of data to extract the main features. EOF analysis was introduced to meteorological problems by Lorenz (1956) and was widely used in meteorology and other fields in the late 1960s (Kutzbach, 1967). The data are represented by EOFs and

temporal coefficients after EOF analysis. The EOFs are the feature vectors, which reflect the spatial information of the data; the temporal coefficients are the PCs, which reflect the weight changes of each spatial mode over time. The specific steps of EOF analysis are as follows:

Step 1. The matrix form of spatiotemporal series data is given as follows:

$$X = \begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1T} \\ X_{21} & X_{22} & \cdots & X_{2T} \\ X_{31} & X_{32} & \cdots & X_{3T} \\ X_{41} & X_{42} & \cdots & X_{4T} \\ X_{51} & X_{52} & \cdots & X_{5T} \end{bmatrix}, \quad (1)$$

where $X_{1t}, \dots, X_{5t}$ ($t = 1, \dots, T$) are the regional data of SWH, MWP, WSPD, SWH1, and SHWW at time t , respectively, and T is the length of the time series. X_{it} ($i = 1, \dots, 5; t = 1, \dots, T$) is a matrix with a size of $n \times 1$, X is a matrix with a size of $5n \times T$, and n is the number of grid points.

Step 2. Calculate the average of the elements of each row of X and denote it as the column vector $\bar{X}$. Let $\bar{X} = (\bar{x}_1, \dots, \bar{x}_n)$, $\bar{X}$ is a matrix with a size of $5n \times T$. X is centralized, as follows:

$$X_1 = X - \bar{X}, \quad (2)$$

where X_1 is the centralized matrix, and the average of the elements of each row of X_1 is 0.

Step 3. Calculate the standard deviation of the elements of each row of X and denote it as the column vector $\tilde{X}$. Let $\tilde{X} = (\tilde{x}_1, \dots, \tilde{x}_n)$, $\tilde{X}$ is a matrix with a size of $5n \times T$. X_1 is normalized, as follows:

$$X_2 = X_1 / \tilde{X}, \quad (3)$$

where $/$ represents the division of the corresponding elements of two matrices, X_2 is the normalized matrix, the average of the elements of each row of X_2 is 0, and the standard deviation is 1.

Step 4. Calculate the product of X_2 and its transpose matrix X_2^T to obtain the correlation coefficient matrix C , as follows:

$$C = \frac{1}{T} X_2 X_2^T, \quad (4)$$

where C is a matrix with a size of $5n \times 5n$.

Step 5. Eigenvalue decomposition is performed on C to obtain the eigenvectors and eigenvalues λ_i ($i = 1, \dots, 5n$), as follows:

$$C = V Q V^T, \quad (5)$$

where Q is a diagonal matrix composed of eigenvalues and V is a matrix composed of eigenvectors. Each eigenvector is a spatial mode.

Step 6. Calculate the principal component P , which is the temporal coefficient corresponding to each spatial mode, as follows:

$$P = V^T X_2, \quad (6)$$

where each row vector of P is the temporal coefficient of the corresponding spatial mode.

Step 7. The contribution rate of each spatial mode is calculated. The variance contribution rate of the k th mode is as follows:

$$\lambda_k = \frac{\lambda_k}{\sum_{i=1}^{5n} \lambda_i} \times 100\%. \quad (7)$$

Step 8. In order to know whether the obtained EOFs are random, a significance test at the 95% confidence level is required. After the test is completed, the modes that do not pass the significance test are removed.

After completing the EOF analysis, X_2 can be represented by EOFs and PCs, as follows:

$$X_2 = VP. \quad (8)$$

The matrix X can be represented as follows:

$$X = VP \odot \tilde{X} + \bar{X}, \quad (9)$$

where $\odot$ is the Hadamard product, which represents the multiplication of the corresponding elements of two matrices.

Multiple EOFs can be obtained by EOF analysis, but as long as the total variance contribution rate is large enough. This method can significantly reduce the data dimensionality and greatly improve the computational efficiency.

3.2. EEMD

The purpose of empirical mode decomposition (EMD) is to decompose the original data into a superposition of multiple independent components, which are called IMFs and residues. Although EMD is widely used in many fields such as speech recognition and signal analysis (Oh et al., 2019), it has a modal aliasing problem. To solve this problem, EEMD was proposed (Wu and Huang, 2009). First, due to the uniform frequency distribution of white Gaussian noise, white Gaussian noise is added to the original data and EMD is performed several times. Then, the corresponding IMFs are averaged to cancel the added white noise, so as to suppress the modal aliasing problem.

IMFs must satisfy the following two conditions:

- (1) In the time range, the difference between the number of zero points and local extreme points of the function is 0 or 1.
- (2) The envelope of local maximum and the envelope of local minimum are symmetric about the time axis.

The specific steps of EEMD are as follows:

Step 1. To obtain a new signal, add the original signal and a white noise obeying a normal distribution, as follows:

$$x_i(t) = x(t) + n_i(t), i = 1, \dots, N, \quad (10)$$

where $x(t)$ is the original signal, $n_i(t)$ is the i th white Gaussian noise, $x_i(t)$ is the i th signal with noise added, and N is the number of new signals.

Step 2. EMD is performed on $x_i(t)$ to obtain M IMFs and a residue. The decomposed $x_i(t)$ is as follows:

$$x_i(t) = \sum_{j=1}^M I_{ij}(t) + R_i(t), \quad (11)$$

where $I_{ij}(t)$ is the j th IMF of $x_i(t)$ after EMD, $R_i(t)$ is the residue of $x_i(t)$ after EMD.

Step 3. Repeat steps 1 and 2 for $x_1(t), \dots, x_N(t)$, respectively, so that M IMFs and one residue are obtained for each signal.

Step 4. Based on the principle that the statistical mean of the uncorrelated series is 0, the mean of the corresponding IMFs and residues are calculated as the IMFs of the EEMD, as follows:

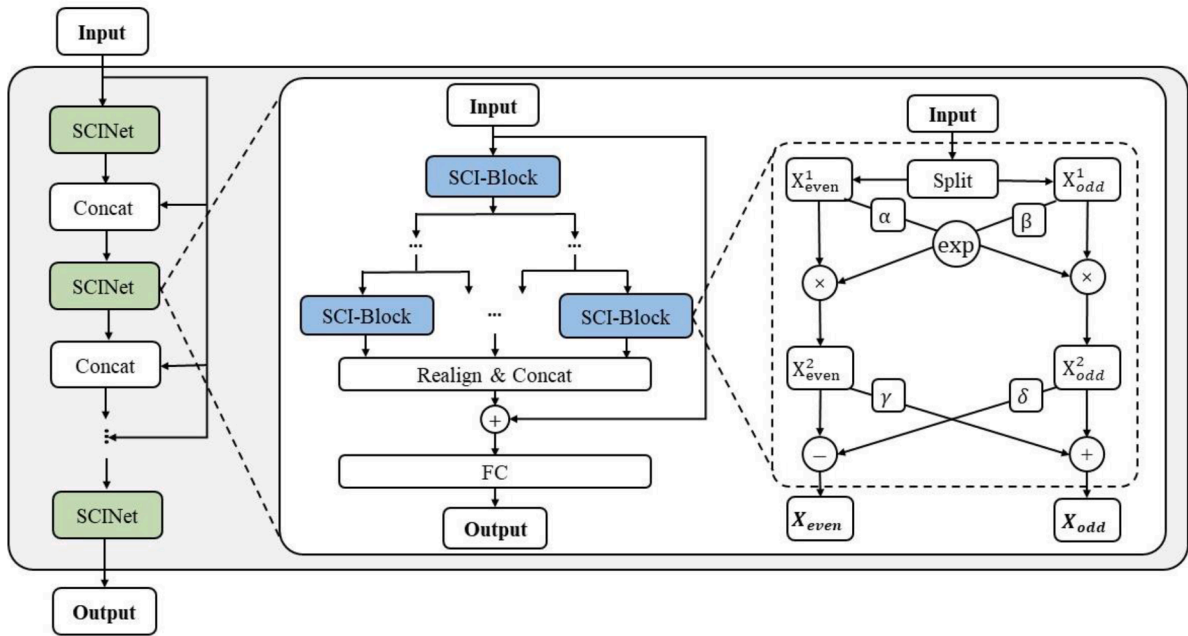


Fig. 3. The architecture of SCINet. The right part represents stacked-SCINet, the middle part represents SCINet and the left part represents SCI-Block.

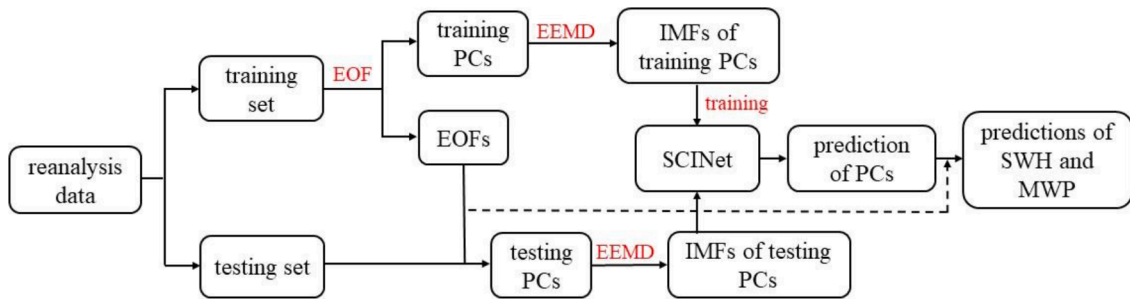


Fig. 4. The flowchart of EOF-EEMD-SCINet.

Table 1
Hyperparameter tuning values.

Parameter	Final value
Batch size	18
Learning rate	6e-5
Dropout	0.3
Level	1
Stack	2
Kernel	7
Hidden size	1
Patience	5

$$I_j(t) = \frac{1}{N} \sum_{i=1}^N I_{ij}(t), j = 1, \dots, M, \quad (12)$$

$$R(t) = \frac{1}{N} \sum_{i=1}^N R_i(t), \quad (13)$$

where $I_j(t)$ is the j th IMF of $x(t)$ after EEMD, and $R(t)$ is the residue of $x(t)$ after EEMD.

After completing EEMD, the original time series can be represented as follows:

$$x(t) = \sum_{j=1}^M I_j(t) + R(t). \quad (14)$$

3.3. SCINet

SCINet is a newly proposed neural network architecture that conducts sample convolution and interaction for temporal modeling. Currently, SCINet has been applied to water quality monitoring (Im et al., 2022) and stock prediction (Yuan et al., 2023). SCINet is a hierarchical time series forecasting framework. The basic building block of this network is SCI-Block, which extracts homogeneous and heterogeneous information from the original data. The interactive learning between the two subsequences compensates for the loss of information. Multiple SCI-Blocks are arranged into a binary tree structure called SCINet, in which the time series obtains effective features with different temporal resolutions. After realignment and adding the original time series, the prediction of the original time series is achieved. To improve the prediction accuracy, multiple SCINets are stacked appropriately to obtain a stacked SCINet.

SCI-Block, as the basic module of SCINet, has the architecture shown in the dashed box on the right of Fig. 3. The specific steps of SCI-Block are as follows:

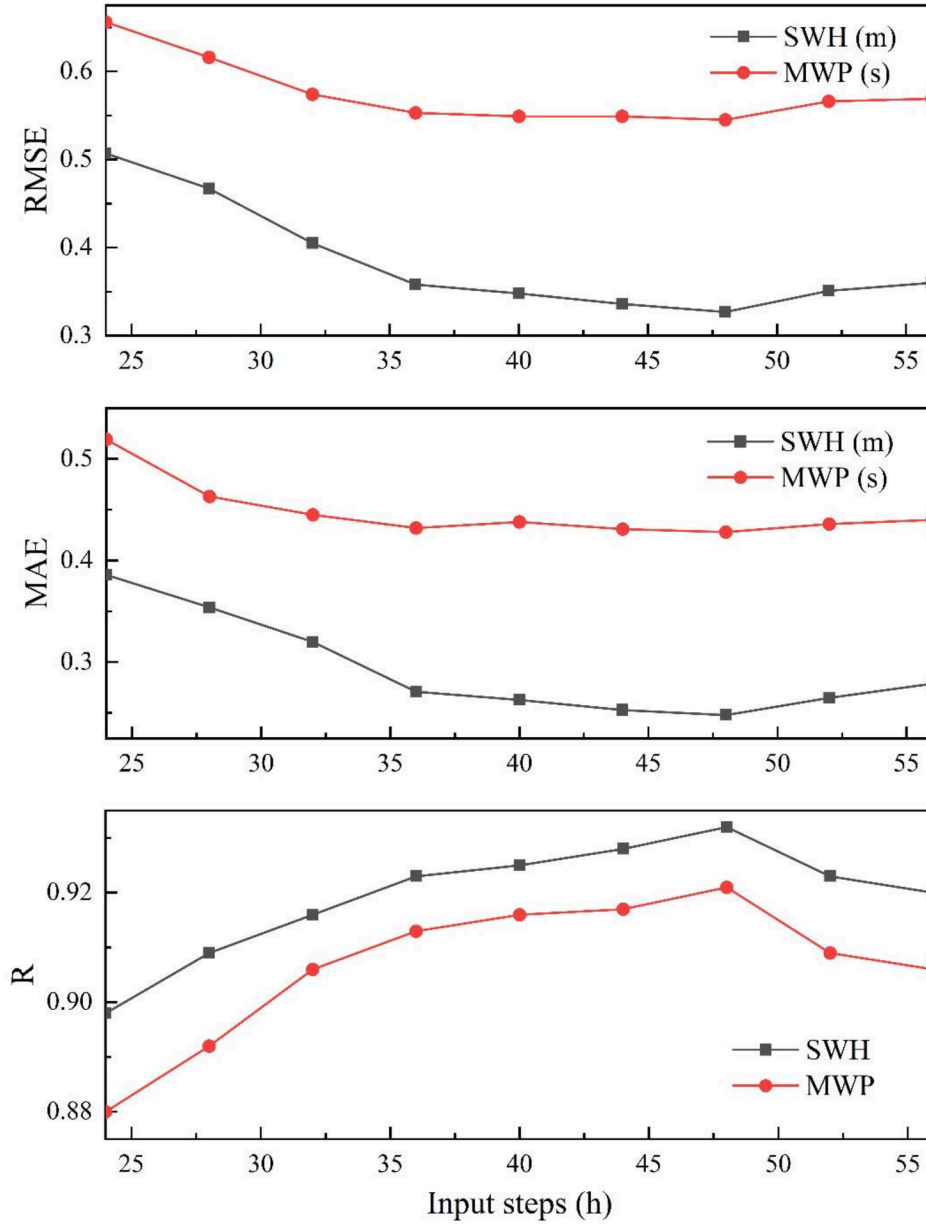


Fig. 5. The RMSE (top), MAE (middle) and R (bottom) of the next 24 h SWH (black) and MWP (red) of different input steps.

Step 1. Two subsequences X_{even}^1 and X_{odd}^1 are obtained by separating the even and odd elements of the original time series X .

Step 2. Two different convolution modules α and β are used to obtain $\alpha(X_{even}^1)$ and $\beta(X_{odd}^1)$, thus extracting features from X_{even}^1 and X_{odd}^1 , respectively.

Step 3. Information is interchanged between the two subsequences through an interactive learning strategy, thereby compensating for the potential information loss caused by downsampling. The first step of the interactive learning is performed, which is the scaling transformation of X_{even}^1 and X_{odd}^1 . $\alpha(X_{even}^1)$ and $\beta(X_{odd}^1)$ obtained from step 2 are converted to the format of exp, and then they are interacted with X_{odd}^1 and X_{even}^1 , respectively, as follows:

$$X_{even}^2 = X_{even}^1 \odot \exp(\beta(X_{odd}^1)), X_{odd}^2 = X_{odd}^1 \odot \exp(\alpha(X_{even}^1)), \quad (15)$$

where $\odot$ is the Hadamard product. $\exp(\beta(X_{odd}^1))$ and $\exp(\alpha(X_{even}^1))$ are regarded as scaling factors.

Step 4. The second step of interactive learning is performed to obtain the final outputs of the SCI-Block, namely X_{even} and X_{odd} . Using two other different convolution modules γ and δ , X_{even}^2 and X_{odd}^2 are projected to the hidden states to obtain $\gamma(X_{even}^2)$ and $\delta(X_{odd}^2)$, which are added to or subtracted from X_{odd}^2 and X_{even}^2 , respectively, as follows:

$$X_{even} = X_{even}^2 \pm \delta(X_{odd}^2), X_{odd} = X_{odd}^2 \pm \gamma(X_{even}^2), \quad (16)$$

where the addition and subtraction operators are learned adaptively by the model during training.

α , β , γ and δ are convolution modules with the same architecture composed of multiple parts such as ReplicationPad1d, Conv1d, LeakyReLU, Dropout and Tanh.

As shown in the middle part of Fig. 3, SCINet is a tree-like structure, which is constructed by hierarchical arrangement of multiple SCI-Blocks. When time series is input into SCINet, effective features with different temporal resolutions are learned using multiple layers of SCI-

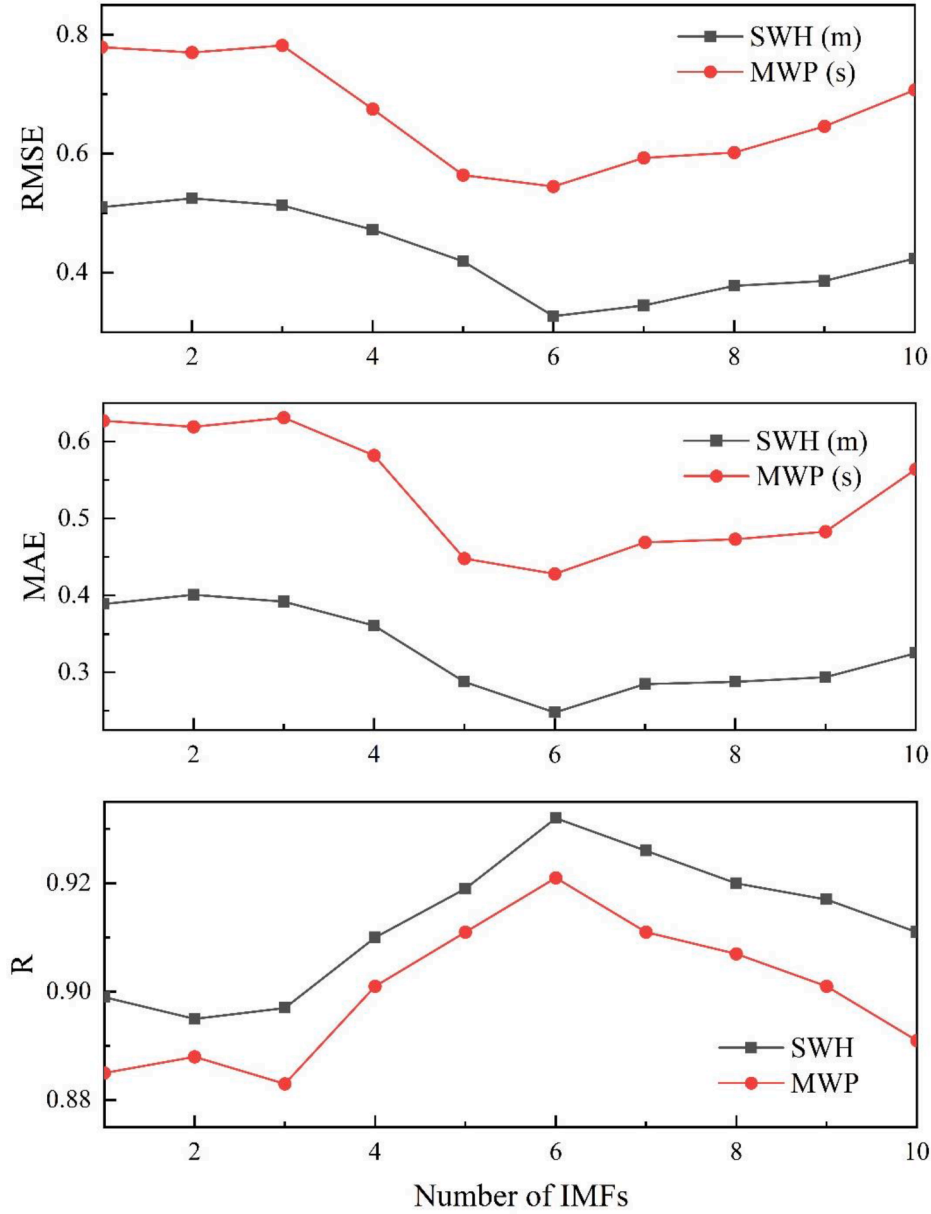


Fig. 6. The RMSE (top), MAE (middle) and R (bottom) of the next 24 h SWH (black) and MWP (red) of different number of IMFs.

Blocks. After the input is processed by all the SCI-Blocks, the elements in the sub-features are rearranged into a new sequence and added to the original time series for prediction. Finally, a fully connected layer is used to obtain the output of SCINet.

In addition to using a single SCINet, multiple layers of SCINets can be stacked, as shown in the left part of Fig. 3. To ease the learning of intermediate temporal features, intermediate supervision is applied to the output of each SCINet. The output of each intermediate SCINet is concatenated with a part of its input in order to recover the length of the original input. It is then transmitted as input to the next SCINet. The output of the last SCINet is the output of the stacked SCINet.

3.4. The proposed model (EOF-EEMD-SCINet)

In order to improve the prediction precision of regional SWH and MWP, a hybrid prediction model combining EOF analysis, EEMD and SCINet is proposed in this study.

The whole framework of the proposed model is shown in Fig. 4. The main steps are as follows:

Step 1. The original reanalysis data are divided into a training set and a test set. The data from January 2020 to September 2020 are the training data and the data from October 2020 to December 2020 are the test data.

Step 2. Multivariate EOF analysis is performed on the training dataset to obtain the EOFs and corresponding PCs of the training dataset. The total variance contribution rate of the nine EOFs obtained is 80% and they all pass the significance test.

Step 3. The EOFs of the training dataset are used to calculate the PCs of the test dataset, as follows:

$$PC_{test} = EOF_{train}^T \cdot X_{test}, \quad (17)$$

where PC_{test} is the PCs of the test dataset, EOF_{train} is the EOFs of the training dataset, and X_{test} is the matrix form of the normalized test data.

Step 4. The PCs of the training dataset and the PCs of the test dataset are decomposed into a set of IMFs by EEMD algorithm, respectively.

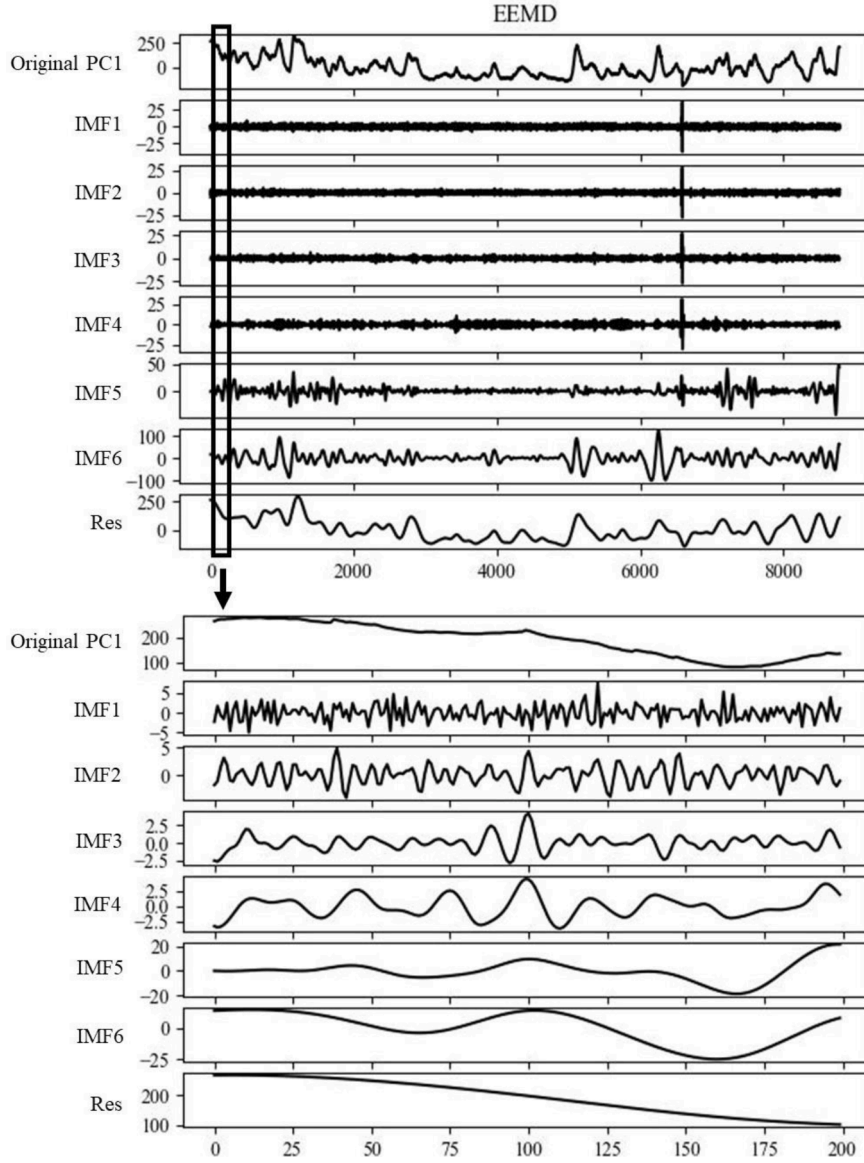


Fig. 7. The decomposed results of PC1 using EEMD, including six IMFs and one residue. The decomposition result of the first 200 data is used as subfigure.

Table 2

24-hour SWH and MWP forecasting results of different decomposition methods. The best result is highlighted in bold.

Methods	SWH RMSE	MAE	R	MWP RMSE	MAE	R
EMD	0.381	0.295	0.916	0.591	0.471	0.910
EEMD	0.327	0.248	0.932	0.545	0.428	0.921
CEEMD	0.373	0.286	0.923	0.581	0.462	0.916
CEEMDAN	0.349	0.265	0.925	0.564	0.446	0.917
EWT	0.378	0.292	0.919	0.585	0.472	0.913
LMD	0.496	0.385	0.895	0.810	0.667	0.884

Step 5. IMFs of training PCs are used to train SCINet. Using the IMFs of testing PCs as input, the PCs of SWH and MWP are predicted by the trained SCINet.

Step 6. The predicted PCs is combined with the EOFs obtained in step 2 to obtain the predicted values of SWH and MWP.

4. Optimization of the model

4.1. Comparison of input steps

For regional forecasting models, the input step affects the performance of the model. The appropriate increase of input data provides more information to the model, which makes the prediction better. However, too long input data contains too much irrelevant information, which leads to lower quality forecasts (Huang and Dong, 2021).

In order to evaluate the predictive effect of the proposed model, three commonly used statistical metrics are used. RMSE, mean absolute error (MAE) and R are used to evaluate the performance of the forecasting models. The statistical metrics are defined as follows:

$$RMSE = \frac{1}{N} \sum_{t=1}^N \sqrt{\frac{1}{M} \sum_{i=1}^M (x_{ti} - y_{ti})^2}, \quad (18)$$

$$MAE = \frac{1}{N} \sum_{t=1}^N \left(\frac{1}{M} \sum_{i=1}^M |x_{ti} - y_{ti}| \right), \quad (19)$$

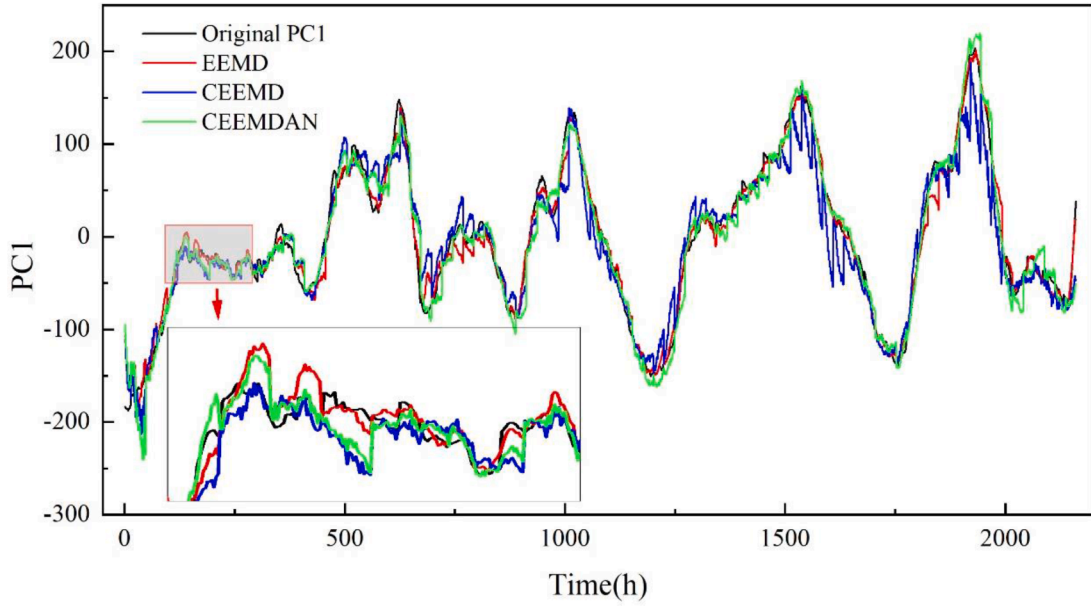


Fig. 8. The forecasting results of PC1 with lead time of 24 h using EEMD (red), CEEMD (blue) and CEEMDAN (green). Black solid line denotes the original PC1. The subfigure is a comparison of the 100th to 300th data.

Table 3

24-hour SWH and MWP forecasting results of different models. The best result is highlighted in bold.

Models	SWH RMSE	MAE	R	MWP RMSE	MAE	R
LSTM	0.447	0.344	0.909	0.668	0.549	0.901
SCINet	0.415	0.317	0.912	0.639	0.519	0.907
EEMD-LSTM	0.391	0.305	0.920	0.636	0.516	0.912
EEMD-SCINet	0.327	0.248	0.932	0.545	0.428	0.921

$$R = \frac{1}{N} \sum_{i=1}^N \frac{\sum_{j=1}^M (x_{ti} - \bar{x}_t)(y_{ti} - \bar{y}_t)}{\sqrt{\sum_{j=1}^M (x_{ti} - \bar{x}_t)^2 \sum_{j=1}^M (y_{ti} - \bar{y}_t)^2}}, \quad (20)$$

where N is the number of field data, M is the number of grid points, x_{ti} is the predicted value, y_{ti} is the target value, $\bar{x}_t$ is the mean predicted value at time t , and $\bar{y}_t$ is the mean target value at time t . As seen in the formula, the average values of RMSE, MAE and R are calculated twice, which is because the data used are three-dimensional. The average value about space is first calculated, and then the average value about time is calculated to obtain the final statistical metrics.

There are many hyperparameters in SCINet that need to be optimized, such as batch size, learning rate and dropout. In order to achieve the best performance of the model, the random search algorithm is used to optimize the hyperparameters in this paper. The performance of random search algorithm is better than grid search and manual search (Bergstra and Bengio, 2012). The number of levels and stacks should not be too much in order to avoid the model being too complex and leading to training difficulties. In most cases, level ≤ 5 and stack ≤ 3 are sufficient. To prevent overfitting, dropout regularization with a dropout value of 0.3 and early stopping with a patience of 5 are employed. In the experiment, the epoch is set to 100, and the MAE loss function and Adam optimizer are used for training. Other hyperparameters obtained by the random search algorithm are shown in Table 1.

The input step is crucial, so a sensitivity analysis of the changes in model performance caused by different input steps is performed in this section. The SWH, MWP, WSPD, SWH1 and SHWW in the region are used as inputs of the proposed model to compare the predicted results of the 24 h SWH and MWP with different input steps. The specific

comparison of the evaluation metrics is shown in Fig. 5.

As shown in Fig. 5, the RMSE and MAE of SWH and MWP show a downward trend with the increase of the time span of input data, which indicates that the forecasting performance of the proposed model is gradually improving. However, this does not mean that the larger the time span, the better the performance of the model. The performance of the prediction model stabilizes when the input step reaches 36 h. After the input step reaches 48 h, the RMSE and MAE of SWH and MWP tend to increase slightly, and the R tends to decrease. This is because too much input information with low correlation influences the forecasting precision of the model. The input step of the model with the best performance is 48 h, the RMSE, MAE and R of SWH are 0.327 m, 0.248 m and 0.932, respectively, and the RMSE, MAE and R of MWP are 0.545 s, 0.428 s and 0.921, respectively. Therefore, the time span of 48 h is selected to further study the performance of the proposed model. Although different time spans correspond to different forecasting errors, all results are within the acceptable level, which shows that the model proposed in this study has not only high precision but also high stability.

4.2. Comparison of the number of IMFs

The performance of the forecasting model is affected not only by the input steps but also by the number of IMFs. The original data can be decomposed into a superposition of different numbers of independent components by EEMD. The number of IMFs is different, so the features extracted from the original data are different, and then the performance of the forecasting model is different. Therefore, a sensitivity analysis of the changes in model performance caused by different numbers of IMFs is necessary.

After EOF analysis, the obtained PCs are decomposed into 1 to 10 IMFs and a residue, respectively, using EEMD. Then, SCINet is used to forecast the next 24 h SWH and MWP, and these results are compared. The specific evaluation metrics are shown in Fig. 6. It should be noted that the number of IMFs shown in the figure does not include the number of residues. As the number of IMFs increases, the RMSE and MAE of SWH and MWP decrease gradually at first, and the R increases gradually at first. This is normal because the more IMFs there are, the more features are obtained from the original data. When the PCs are decomposed into six IMFs and one residue, the predictions of SWH and MWP are the best. The performance of the forecasting model becomes poorer when there

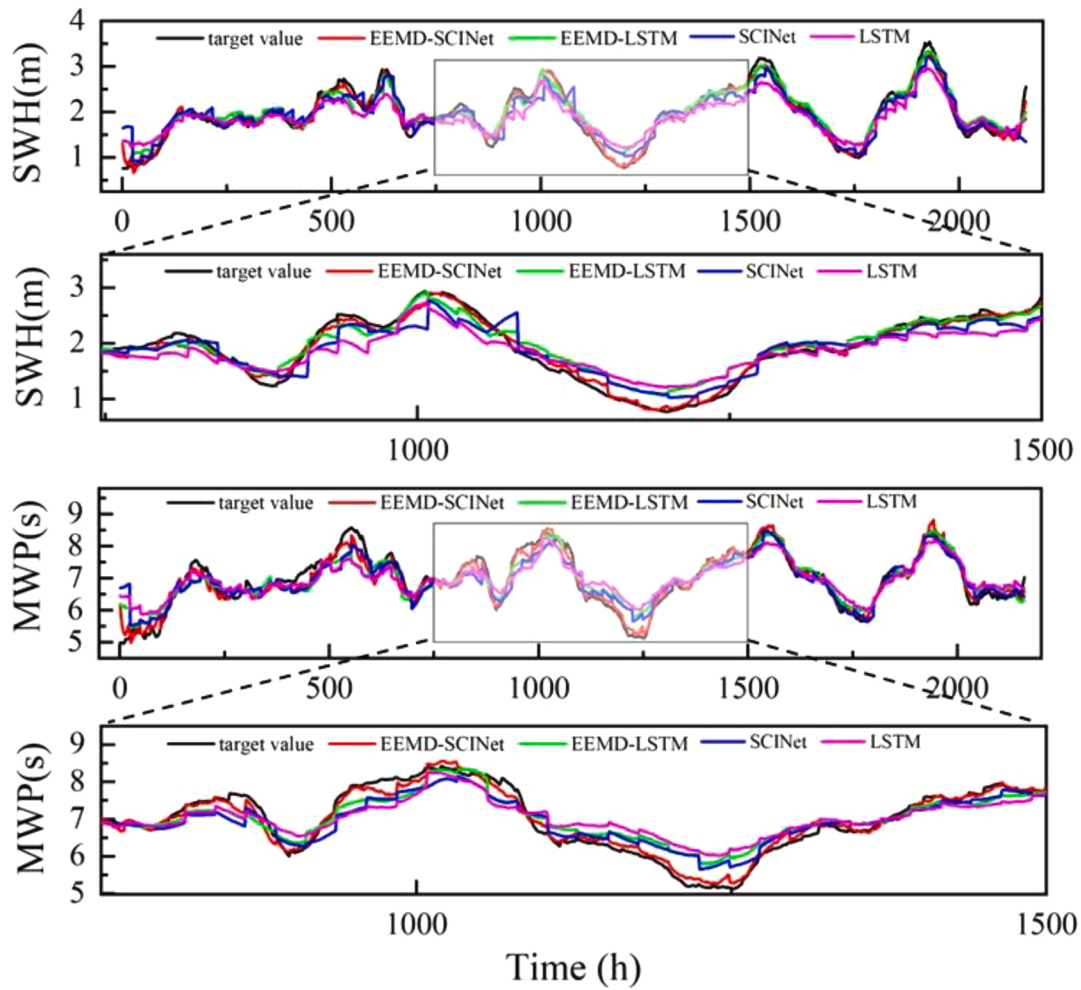


Fig. 9. Spatially averaged SWH and MWP predictions with lead time of 24 h using EEMD-SCINet (red), EEMD-LSTM (green), SCINet (blue) and LSTM (purple). Black solid line denotes the target value. The subplots from top to bottom correspond to SWH, SWH at the 750th to 1500th moments, MWP, and MWP at the 750th to 1500th moments, respectively.

are more IMFs, which shows that the more features extracted is not better. Therefore, decomposing the PCs into six IMFs and one residue is the best choice in this study.

Fig. 7 shows the decomposition result of the first PC in PCs by EEMD, and PC1 is decomposed into six IMFs and one residue. In order to show the trend of the series more clearly, the decomposition result of the first 200 data is used as subfigure. The horizontal axis represents the number of data, and the vertical axis represents the value of each independent component. What is shown in the figure is that the characteristics of each IMF and residue are different. From IMF1 to IMF6 to the residue, the complexity and frequency of the series gradually decrease, and the period gradually increases. The trends of the first four IMFs are the most complex and variable, and their values are relatively much smaller. The changes of IMF5 and IMF6 are more intuitive, and the range is slightly larger. Although the value of residue is the largest, its volatility is the smallest, which reflects the overall trend of PC1. The trend and change pattern of the decomposed series are more obvious than those of the original series, so the decomposition algorithm makes it easier for SCINet to learn the potential mapping between the input and output. In addition, the residue obtained is the intermediate residue that can still be decomposed, rather than the final residue obtained after complete decomposition, because the number of IMFs is preset (Meng et al., 2022). Therefore, the residue obtained is not monotonic.

4.3. Comparison of decomposition methods

The idea of each decomposition algorithm is different, and not all decomposition algorithms are suitable for the model proposed in this study. In order to select the most suitable decomposition algorithm for the proposed forecasting model, six decomposition algorithms are compared in this section, including EMD, EEMD, complementary ensemble empirical modal decomposition (CEEMD) (Yeh et al., 2010), complete EEMD with adaptive noise (CEEMDAN) (Torres et al., 2011), empirical wavelet transform (EWT) (Gilles, 2013), and local mean decomposition (LMD) (Smith, 2005). EMD and EEMD are introduced in Section 3.2. EEMD, CEEMD and CEEMDAN are all adaptive signal decomposition methods developed from EMD. EWT combines the advantages of EMD and Wavelet Transform and provides a new idea for signal decomposition. Like EMD, LMD is also an adaptive signal processing method, but the difference is that LMD avoids the Hilbert transform (Emeksiz and Tan, 2022).

After decomposing the PCs obtained from the EOF analysis into six IMFs and one residue using these six decomposition algorithms, the forecasting of the next 24 h SWH and MWP is performed using SCINet. Table 1 shows the specific settings of the hyperparameters. The results of each decomposition algorithm are shown in Table 2. It is clear that using EEMD to predict SWH and MWP is better than using the other five decomposition methods, and EEMD is the most suitable decomposition method for the model proposed in this study. Although the results of

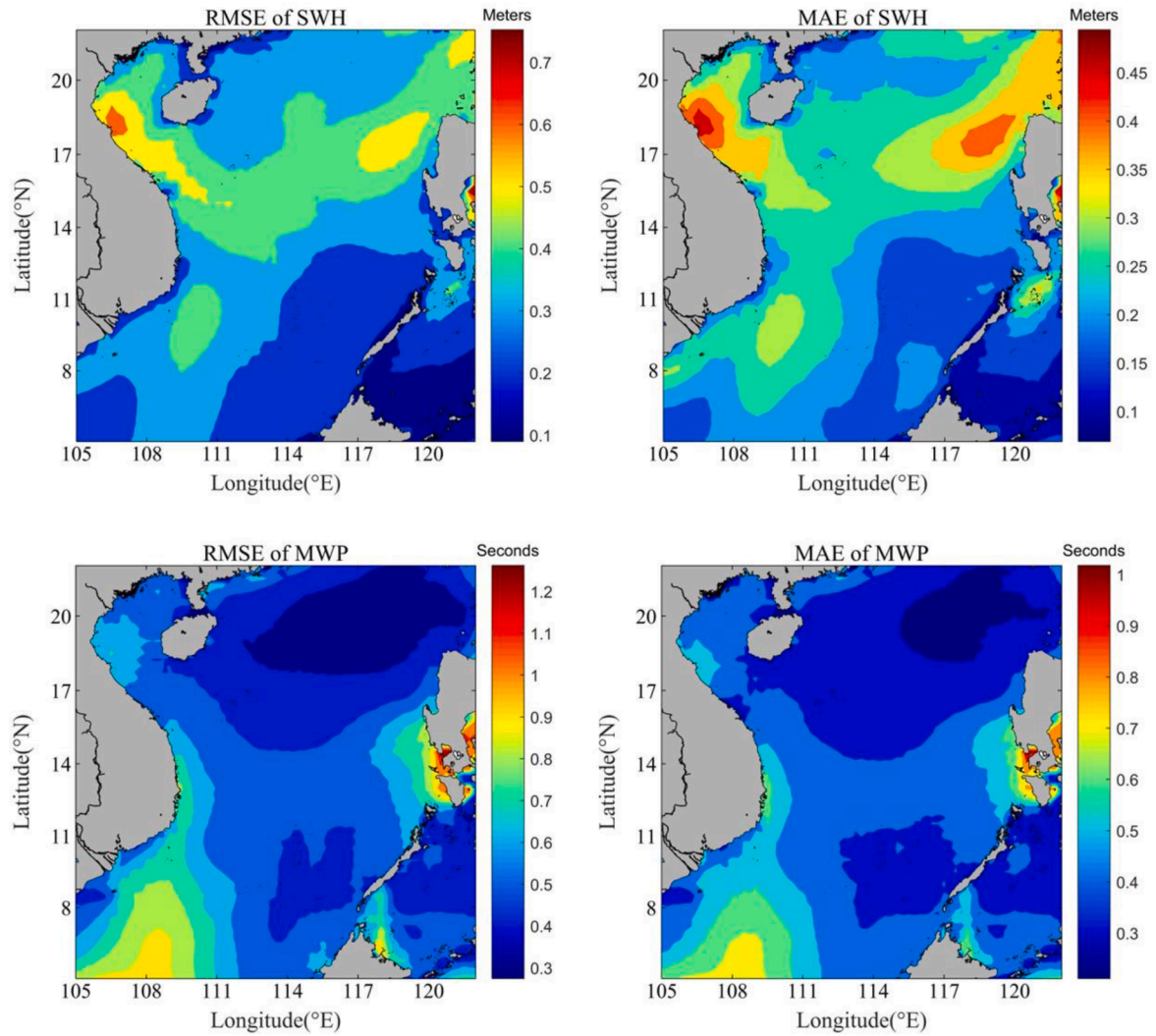


Fig. 10. Distribution of error metrics of 24-hour SWH and MWP predicted using EOF-EEMD-SCINet. The top left, top right, bottom left, and bottom right subplots correspond to RMSE of SWH, MAE of SWH, RMSE of MWP, and MAE of MWP, respectively.

Table 4

48-hour and 72-hour SWH and MWP forecasting results of different models. The best results are highlighted in bold.

Lead time	Models	SWH			MWP		
		RMSE	MAE	R	RMSE	MAE	R
48h	LSTM	0.554	0.440	0.892	0.708	0.580	0.885
	SCINet	0.473	0.365	0.907	0.676	0.552	0.901
	EEMD-LSTM	0.470	0.361	0.910	0.675	0.551	0.903
	EEMD-SCINet	0.355	0.269	0.923	0.562	0.444	0.918
	EEMD-SCINet	0.355	0.269	0.923	0.562	0.444	0.918
72h	LSTM	0.649	0.512	0.885	0.831	0.694	0.874
	SCINet	0.550	0.429	0.893	0.759	0.627	0.889
	EEMD-LSTM	0.525	0.404	0.901	0.726	0.596	0.896
	EEMD-SCINet	0.399	0.305	0.919	0.586	0.469	0.917
	EEMD-SCINet	0.399	0.305	0.919	0.586	0.469	0.917

CEEMDAN are not as good as those of EEMD, CEEMDAN is also a good decomposition method. In addition, the performance of LMD is the worst, and the gap with other methods is large.

Fig. 8 shows the predicted values of PC1 with lead time of 24 h using EEMD, CEEMD and CEEMDAN, and the target values are also plotted for reference. In Table 2, the forecasting results of these three methods are

the top three. As shown in Fig. 8, the prediction curves of the three methods do not differ much, and all of them can roughly capture the trend of PC1. The subfigure is a comparison of the 100th to 300th data, which shows the specific differences of the three prediction curves. It can be seen that the predicted curve of EEMD is generally consistent with the target curve, despite the deviations at individual peaks. These deviations are acceptable due to the high volatility of PC1. Furthermore, the prediction curves of CEEMD and CEEMDAN do not capture the subtle trends of PC1 and are in poor agreement with the target curve. In summary, the performance of EEMD is the best in this section.

5. Results and discussion

5.1. 24-hour SWH and MWP forecasts

The purpose of this section is to evaluate the forecasting performance of EOF-EEMD-SCINet for 24-hour SWH and MWP. EEMD-SCINet is compared with SCINet, LSTM, and EEMD-LSTM using the reanalysis data after EOF analysis. The SCINet model plays the role of the ablation experiment.

The 24-hour SWH and MWP forecasting results of the four models are presented in Table 3. It is obvious from the results that EEMD-SCINet significantly outperforms the other models, whether forecasting SWH or MWP. The RMSE, MAE and R of SWH predicted using EEMD-SCINet

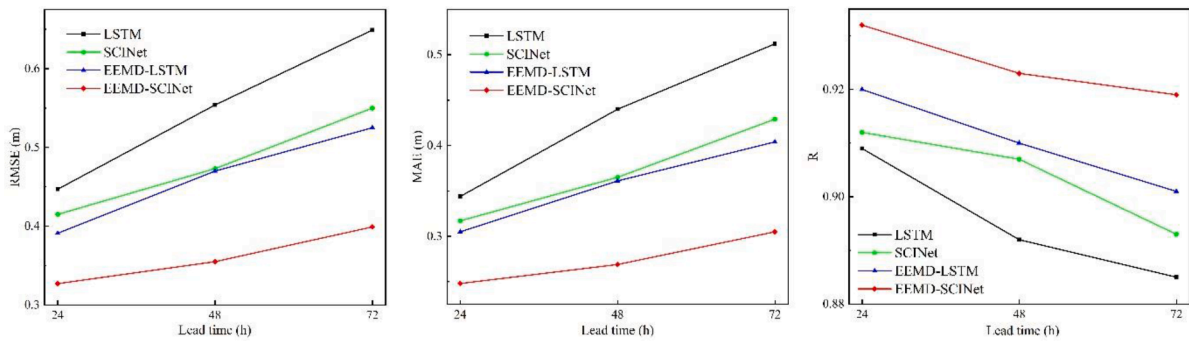


Fig. 11. The forecasting performance of LSTM (black), SCINet (green), EEMD-LSTM (blue) and EEMD-SCINet (red) for SWH with different lead times. The left, middle and right subplots correspond to RMSE, MAE and R, respectively.

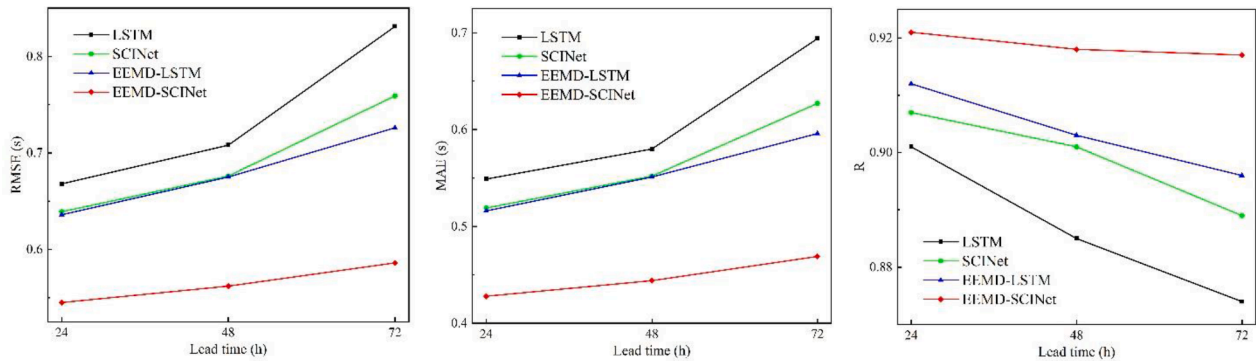


Fig. 12. The forecasting performance of LSTM (black), SCINet (green), EEMD-LSTM (blue) and EEMD-SCINet (red) for MWP with different lead times. The left, middle and right subplots correspond to RMSE, MAE and R, respectively.

are 0.327 m, 0.248 m and 0.932, respectively, and the RMSE, MAE and R of MWP are 0.545 s, 0.428 m and 0.921, respectively. The forecasting precision of SWH is generally higher than that of MWP because the value of MWP is larger and more difficult to forecast. Table 3 also shows that the prediction performance of the hybrid model using the decomposition method is better than that of the single model. This confirms that the decomposition method enables the forecasting model to obtain more relevant information, so that the performance of the forecasting model is improved. The results show that the forecasting precision of SCINet can be significantly improved by combining EEMD. Furthermore, the comparison of the two single models shows that the forecasting precision of SCINet for SWH and MWP is higher than that of LSTM.

The forecasting results of the 24-hour spatially averaged SWH and MWP using EOF-EEMD-SCINet are shown in Fig. 9. The target values from the ERA5 data and the predictions of the other three models are also plotted in Fig. 9 for reference in the proposed model. The SWH and MWP of the 750th to 1500th moments are used as two subfigures. It can be seen that all four models can roughly capture the trends and fluctuations of SWH and MWP. In most cases, the forecasting results of large and small values by a single model are poor, which indicates that it is difficult for a single model to capture extreme events, and precise prediction of extreme events is very important. The predictions of the proposed model at the peak are in good agreement with the target values. Even when predicting MWP, the data can be reproduced with high precision by the proposed model. This indicates that the forecasting performance of the proposed model is improved by using EEMD, and therefore the prediction of extreme events by the proposed model is more stable and reliable.

To demonstrate the prediction performance of the proposed model in the whole study area more concretely, Fig. 10 shows the spatial distribution of the error metrics of predicted SWH and MWP. It can be seen from the four subfigures that the forecasting precision is relatively high

in most regions, with RMSE and MAE less than 0.5 m and 0.35 m for SWH and less than 0.7 s and 0.5 s for MWP, respectively. The error metrics are all within acceptable ranges compared with the target values. The distribution characteristics of error metrics of SWH and MWP are different. The smaller errors of SWH are concentrated in the southeast of the study area, while the errors of SWH are larger in the northeast and northwest. The northeastern part of the study area is the Luzon Strait and its adjacent waters. The Luzon Strait is an important channel for water exchange between the SCS and the Pacific Ocean, and its flow conditions are very complex and SWH is large, so the forecasting precision of this region is low. Another area with large errors of SWH is Beibu Gulf, because this area is located on the edge and the relevant information extracted here is insufficient. Fig. 10 shows that the forecasting results of MWP in the northeast of the study area are better, while the prediction accuracy in the central-eastern and southwestern parts is lower. The central-eastern part of the study area is located in the southwestern waters of Luzon Island, which is an inshore area with shallow water depth and vulnerable to weather conditions. For the southwest, the prediction result is poor due to the influence of boundary factors and monsoon.

5.2. 48-hour and 72-hour SWH and MWP forecasts

With the development of the field of ocean engineering, precise forecasting of SWH and MWP with long lead time becomes more and more important. However, the longer the lead time of the forecasting, the more difficult the forecasting is and the higher the performance requirements of the proposed model. In this section, 48-hour and 72-hour SWH and MWP of the study area are predicted, and the prediction performance of the proposed model is further analyzed by comparing it with other three models.

Table 4 shows the forecasting results of the four models for SWH and

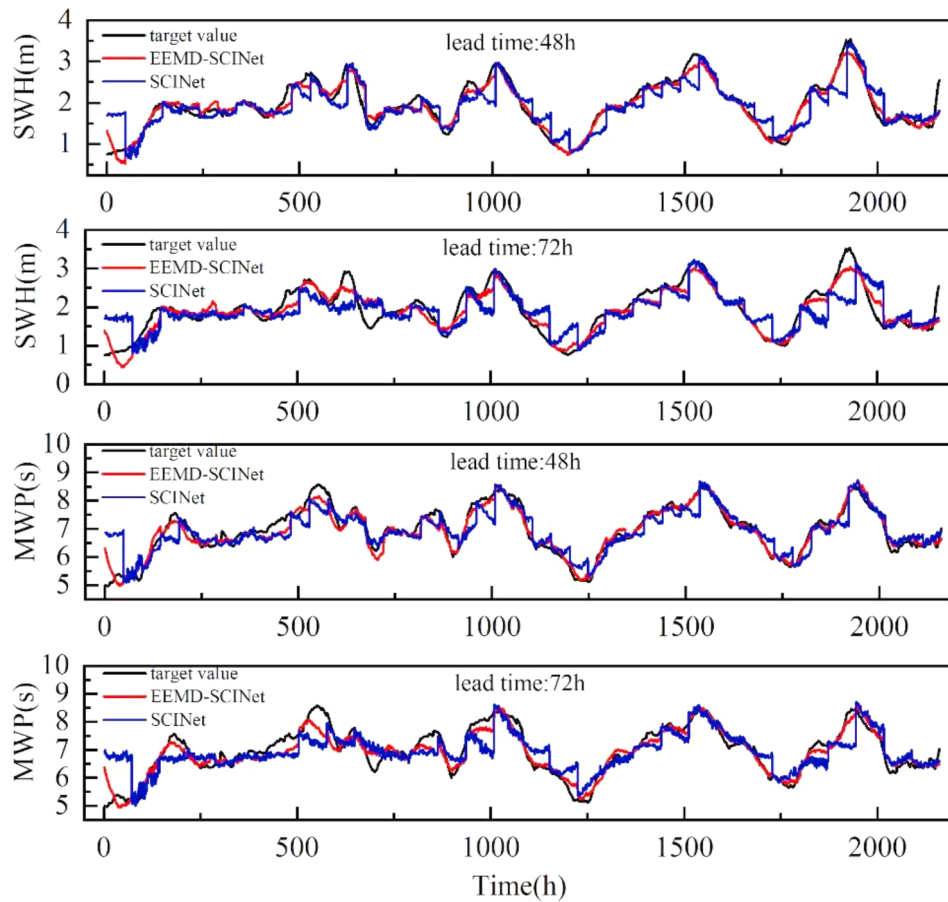


Fig. 13. Spatially averaged SWH and MWP predictions with lead times of 48 h and 72 h using EEMD-SCINet (red) and SCINet (blue). Black solid line denotes the target value.

MWP with different lead times. In 48-hour and 72-hour predictions, the forecasting precision of EEMD-SCINet for SWH and MWP is the highest. Wave conditions in a long-time duration are very variable, making forecasts difficult. As the forecasting time increases, the forecasting error of each model becomes larger. However, the error of EEMD-SCINet increases more slowly compared with other models. Even when forecasting the next 72-hour SWH and MWP, EEMD-SCINet still presents results with high precision. LSTM performs worst at forecasting the next 72-hour SWH and MWP, which indicates the difficulty of long lead time forecast by a single model. In addition, both hybrid models are more precise than the two single models and the error increases more slowly. This fully demonstrates that the forecasting performance of the hybrid model is improved by the decomposition method, especially for longer lead times.

Figs. 11 and 12 intuitively show the forecasting results of different models for SWH and MWP with different lead times. Their three sub-figures correspond to RMSE, MAE and R with different lead times. It can be seen that EEMD-SCINet has the lowest RMSE, the lowest MAE and the highest R at all lead times compared with other three models. With the increase of lead time, the predicting effect of each model for SWH and MWP attenuates, and the gap of statistical metrics among the models gradually widens. The prediction effect of LSTM is seriously attenuated. Moreover, the forecasting results of SCINet and EEMD-LSTM are similar at different lead times, and the performance of EEMD-SCINet is the most stable.

Fig. 13 presents the target and forecasting curves of the spatially averaged SWH and MWP for different lead times using the two forecasting models. When forecasting the next 48-hour SWH and MWP, the forecasting curves of the hybrid model and the target curves are very consistent, and the performance of the model is good even when the

MWP is long. However, the single model performs poorly, with large deviations at many moments, and does not provide reliable predictions of SWH and MWP. For the 72-hour case, the forecasting performance of the hybrid model decreases, and the forecasting precision of a few extreme values is insufficient, but the overall prediction performance is still satisfactory. The forecasting curve of the single model with lead time of 72 h is not only deviated greatly, but also shows abnormal fluctuations. To sum up, the forecasting performance of the model proposed in this study is very excellent.

The regional forecasting results of SWH and MWP with 48-hour and 72-hour lead times using the proposed model are shown in Figs. 14 and 15. Although the 48-hour and 72-hour predictions are slightly less accurate than the 24-hour predictions, they are still acceptable. In the 72-hour forecast, the RMSE and MAE of SWH are concentrated around 0.4 m and 0.25 m, and those of MWP are concentrated around 0.6 s and 0.45 s, respectively. The distribution characteristics of the error metrics are similar to those of the 24-hour results, but the accuracy is slightly lower in individual regions.

5.3. Computing time of the models

The computing time of the four models with different lead times is analyzed in this section. All the experiments have been run on the same computer, using an Intel (R) Core (TM) i7-10750H CPU @ 2.60 GHz processor. The four models are based on EOF analysis, and the running time of EOF analysis is 697 s. The computing time of LSTM and SCINet is mainly the training time of the model, and the computing time of EEMD-LSTM and EEMD-SCINet mainly consists of the decomposition time of the data and the training time of the model. The specific computing time of the four models is shown in Table 5. As the lead time increases, the

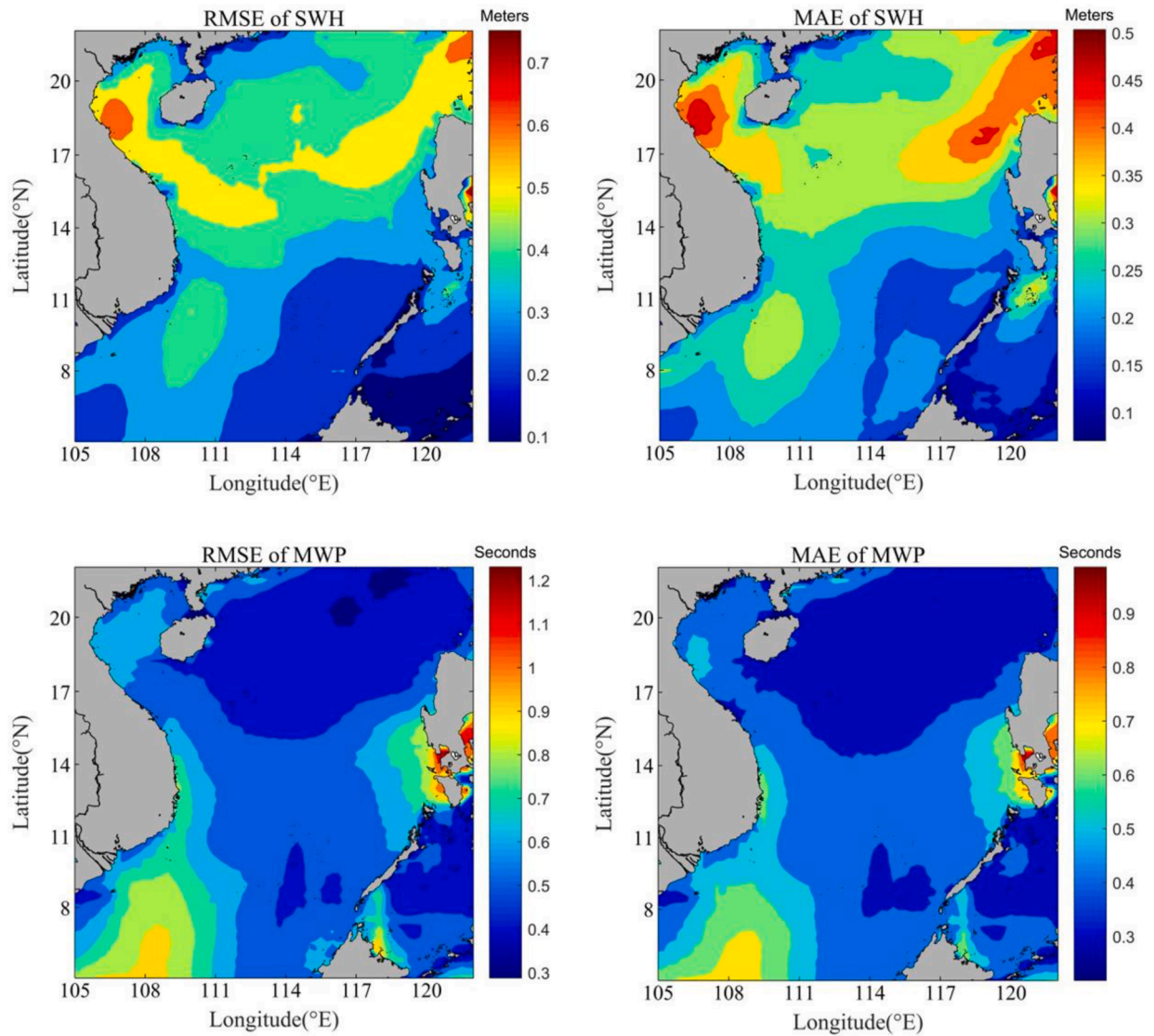


Fig. 14. Distribution of error metrics of 48-hour SWH and MWP predicted using EOF-EEMD-SCINet. The top left, top right, bottom left, and bottom right subplots correspond to RMSE of SWH, MAE of SWH, RMSE of MWP, and MAE of MWP, respectively.

training time of all models becomes longer. Nevertheless, the training time of SCINet and EEMD-SCINet raises slowly compared with LSTM and EEMD-LSTM. Although the forecasting performance of the hybrid model is better than the corresponding single model, the training time of the hybrid model is longer due to the larger amount of data for training the hybrid model. In 24-hour prediction, it is obvious that the training speed of EEMD-SCINet is slower than that of EEMD-LSTM. However, the longer the lead time, the greater the advantage of EEMD-SCINet in terms of training time. The training cost of EEMD-SCINet is approximately half of that of EEMD-LSTM in the 72-hour prediction.

6. Conclusions

In this paper, EOF-EEMD-SCINet model for regional SWH and MWP forecasts is proposed by combining EOF analysis, EEMD and SCINet. EOF analysis is first performed on the reanalysis data of multiple variables to obtain corresponding EOFs and PCs, thus reducing the dimensionality of the data. To extract the complex features in the data, EEMD is performed on the PCs. After that, the decomposed PCs are used as input to forecast the PCs of SWH and MWP using SCINet. Finally, the predicted PCs and EOFs are combined to obtain the final multi-hour regional SWH and MWP.

The SCS is selected as the study area, and the dataset is provided by ECMWF. The sensitivity analysis of the changes in model performance caused by different input steps and number of IMFs is performed to determine the optimal input steps and number of IMFs. The comparison experiment of six decomposition methods shows that EEMD improves the performance of the proposed model to the greatest extent. The performance of the forecasting model is evaluated by using RMSE, MAE and R. The inputs considered include SWH, MWP, WSPD, SWH1 and SHWW. The forecasting performance of the model for regional SWH and MWP with lead times of 24, 48 and 72 h is validated and compared with LSTM, SCINet and EEMD-SCINet based on EOF analysis. From the perspective of time and space, the proposed model can not only accurately forecast the changes in SWH and MWP over time, but also has a high forecasting precision for regional waves, which is superior to other models. The increase of wave lead time from 24 to 72 h has less negative effect on the forecasting performance of the proposed model compared to other models. The MAE of SWH increases by less than 6 cm, and the MAE of MWP changes around 0.04 s. Meanwhile, EOF-EEMD-SCINet has enormous advantage in terms of efficiency compared with EOF-EEMD-LSTM. These results show that EOF-EEMD-SCINet can simultaneously forecast regional SWH and MWP with high precision.

The proposed model is a promising tool for regional wave forecast

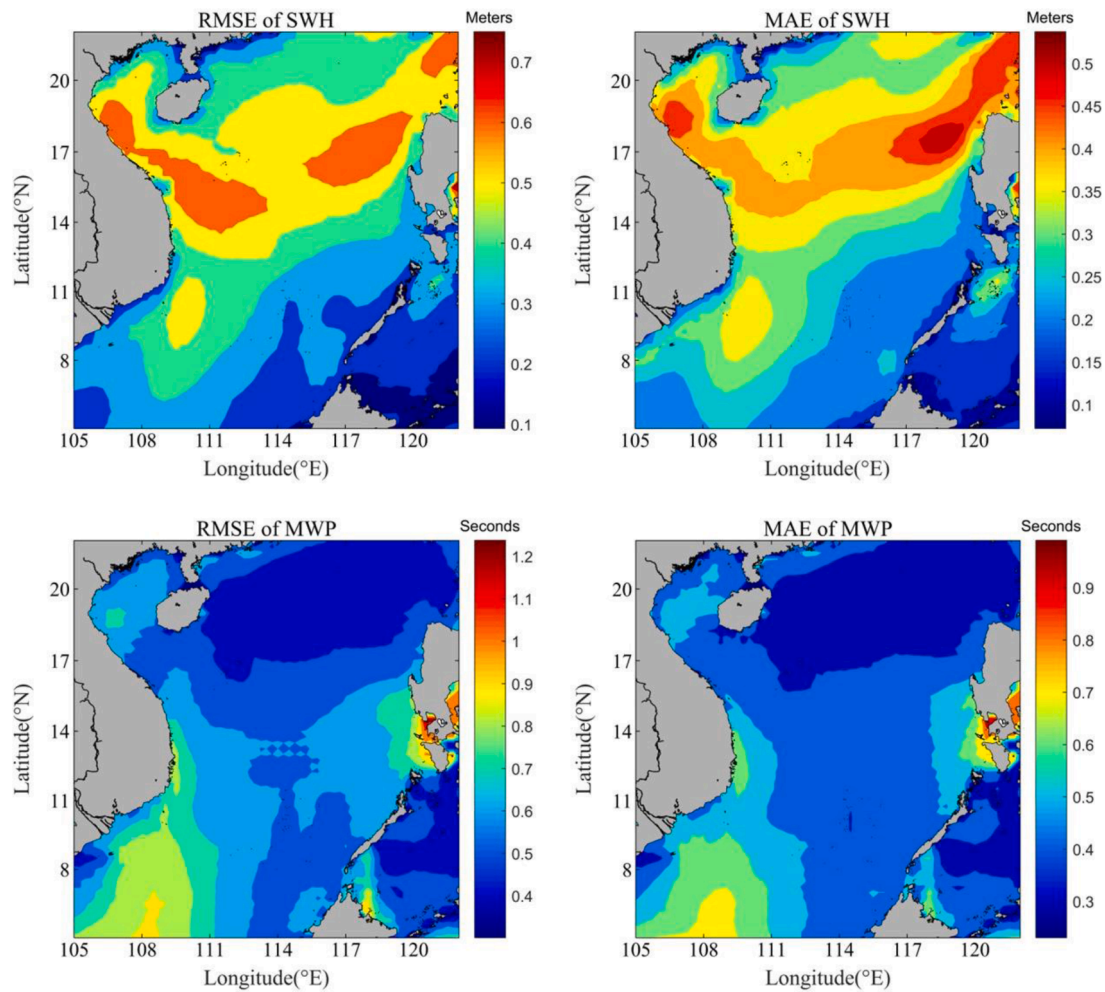


Fig. 15. Distribution of error metrics of 72-hour SWH and MWP predicted using EOF-EEMD-SCINet. The top left, top right, bottom left, and bottom right subplots correspond to RMSE of SWH, MAE of SWH, RMSE of MWP, and MAE of MWP, respectively.

Table 5

Computing times of four models with different lead times.

Models	Decomposing time (s)	Training time (s)		
		24 h	48 h	72 h
LSTM	–	385	863	1014
SCINet	–	150	165	245
EEMD-LSTM	275	642	1236	1412
EEMD-SCINet	275	664	711	787

and has great application value. For future work, spatiotemporal series forecasting models such as Self-Attention ConvLSTM (Lin et al., 2020) and DGGAN (Ying et al., 2018) can be introduced to further validate the superiority of EOF-EEMD-SCINet. The reasons for the superior performance of EEMD over other decomposition methods are not analyzed and discussed in this paper, and further research on these is worthwhile. It should also be noted that the data used in this study are one year of ERA5 data, and the proposed model should be applied to more than one year of data and radar data (Huang et al., 2017; Wyatt et al., 1999) in the future. In addition, deep neural networks have the black-box nature, and numerous scholars have been studying the interpretability of deep neural networks in recent years (Fan et al., 2020a; Zhang et al., 2022). The interpretability of the model can be further studied to identify the potential weaknesses of the model, thus improving its precision and stability.

CRediT authorship contribution statement

Jie Ding: Methodology, Software, Validation, Visualization, Formal analysis, Writing – original draft, Writing – review & editing. **Fangyu Deng:** Conceptualization, Validation, Formal analysis, Writing – original draft, Writing – review & editing. **Qi Liu:** Data curation, Resources. **Jichao Wang:** Conceptualization, Funding acquisition, Investigation, Project administration, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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References

- Agrawal, J., Deo, M., 2002. On-line wave prediction. *Mar. Struct.* 15 (1), 57–74.
- Bai, G., Wang, Z., Zhu, X., Feng, Y., 2022. Development of a 2-D deep learning regional wave field forecast model based on convolutional neural network and the application in South China Sea. *Appl. Ocean Res.* 118, 103012.
- Bergstra, J., Bengio, Y., 2012. Random search for hyper-parameter optimization. *J. Mach. Learn. Res.* 13 (2).
- Booij, N., Ris, R.C., Holthuijsen, L.H., 1999. A third-generation wave model for coastal regions: 1. Model description and validation. *J. Geophys. Res.: Oceans* 104 (C4), 7649–7666.
- Campos, R.M., Krasnopolsky, V., Alves, J.-H., Penny, S.G., 2020. Improving NCEP's global-scale wave ensemble averages using neural networks. *Ocean Modell.* 149, 101617.
- Chang, H.-K., Liou, J.-C., Liu, S.-J., Liaw, S.-R., 2011. Simulated wave-driven ANN model for typhoon waves. *Adv. Eng. Software* 42 (1–2), 25–34.
- Choi, H., Park, M., Son, G., Jeong, J., Park, J., Mo, K., Kang, P., 2020. Real-time significant wave height estimation from raw ocean images based on 2D and 3D deep neural networks. *Ocean Eng.* 201, 107129.
- Cuadra, L., Salcedo-Sanz, S., Nieto-Borge, J., Alexandre, E., Rodríguez, G., 2016. Computational intelligence in wave energy: comprehensive review and case study. *Renew. Sustain. Energy Rev.* 58, 1223–1246.
- Emeksz, C., Tan, M., 2022. Multi-step wind speed forecasting and Hurst analysis using novel hybrid secondary decomposition approach. *Energy* 238, 121764.
- Fan, F., Xiong, J., Li, M., Wang, G., 2020a. On Interpretability of Artificial Neural Networks: a Survey.
- Fan, S., Xiao, N., Dong, S., 2020b. A novel model to predict significant wave height based on long short-term memory network. *Ocean Eng.* 205, 107298.
- Fusco, F., 2009. Short-term wave forecasting as a univariate time series problem. no. December.
- Gilles, J., 2013. Empirical wavelet transform. *IEEE Trans. Signal Process.* 61 (16), 3999–4010.
- Group, T.W., 1988. The WAM model—A third generation ocean wave prediction model. *J. Phys. Oceanogr.* 18 (12), 1775–1810.
- Hu, H., van der Westhuysen, A.J., Chu, P., Fujisaki-Manome, A., 2021. Predicting Lake Erie wave heights and periods using XGBoost and LSTM. *Ocean Modell.* 164, 101832.
- Huang, W., Dong, S., 2021. Improved short-term prediction of significant wave height by decomposing deterministic and stochastic components. *Renew. Energy* 177, 743–758.
- Huang, W., Liu, X., Gill, E.W., 2017. Ocean wind and wave measurements using X-band marine radar: a comprehensive review. *Remote Sens. (Basel)* 9 (12), 1261.
- Im, Y., Song, G., Lee, J., Cho, M., 2022. Deep learning methods for predicting tap-water quality time series in South Korea. *Water (Basel)* 14 (22), 3766.
- Jain, P., Deo, M., 2006. Neural networks in ocean engineering. *Ship. Offshor. Struct.* 1 (1), 25–35.
- Jain, P., Deo, M., 2007. Real-time wave forecasts off the western Indian coast. *Appl. Ocean Res.* 29 (1–2), 72–79.
- James, S.C., Zhang, Y., O'Donncha, F., 2018. A machine learning framework to forecast wave conditions. *Coast. Eng.* 137, 1–10.
- Jing, Y., Zhang, L., Hao, W., Huang, L., 2022. Numerical study of a CNN-based model for regional wave prediction. *Ocean Eng.* 255, 111400.
- Kutzbach, J.E., 1967. Empirical eigenvectors of sea-level pressure, surface temperature and precipitation complexes over North America. *J. Appl. Meteorol. Climatol.* 6 (5), 791–802.
- Lin, Z., Li, M., Zheng, Z., Cheng, Y., Yuan, C., 2020. Self-attention convlstm for spatiotemporal prediction. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 11531–11538.
- Liu, M., Zeng, A., Xu, Z., Lai, Q., Xu, Q., 2021. Time series is a special sequence: forecasting with sample convolution and interaction. *arXiv:2106.09305*.
- Lorenz, E.N., 1956. *Empirical Orthogonal Functions and Statistical Weather Prediction*. Massachusetts Institute of Technology, Department of Meteorology Cambridge.
- Malekmohamadi, I., Bazargan-Lari, M.R., Kerachian, R., Nikoo, M.R., Fallahnia, M., 2011. Evaluating the efficacy of SVMs, BNs, ANNs and ANFIS in wave height prediction. *Ocean Eng.* 38 (2–3), 487–497.
- Meng, F., Xu, D., Song, T., 2022. ATDNNs: an adaptive time–frequency decomposition neural network-based system for tropical cyclone wave height real-time forecasting. *Future Gener. Comput. Syst.* 133, 297–306.
- Minuzzi, F.C., Farina, L., 2023. A deep learning approach to predict significant wave height using long short-term memory. *Ocean Modell.* 181, 102151.
- Oh, B.S., Zhuang, H., Toh, K.A., Lin, Z., 2019. Augmented EMD for complex-valued univariate signals. *IET Signal Proc.* 13 (4), 424–433.
- Reikard, G., 2009. Forecasting ocean wave energy: tests of time-series models. *Ocean Eng.* 36 (5), 348–356.
- Shao, Q., Hou, G., Li, W., Han, G., Liang, K., Bai, Y., 2021. Ocean reanalysis data-driven deep learning forecast for sea surface multivariate in the South China Sea. *Earth Space Sci.* 8 (7), e2020EA001558.
- Smith, J.S., 2005. The local mean decomposition and its application to EEG perception data. *J. R. Soc., Interface* 2 (5), 443–454.
- Sreelakshmi, S., Bhaskaran, P.K., 2020. Wind-generated wave climate variability in the Indian Ocean using ERA-5 dataset. *Ocean Eng.* 209, 107486.
- Stopa, J.E., Cheung, K.F., Tolman, H.L., Chawla, A., 2013. Patterns and cycles in the climate forecast system reanalysis wind and wave data. *Ocean Modell.* 70, 207–220.
- Tolman, H.L., Balasubramanian, B., Burroughs, L.D., Chalikov, D.V., Chao, Y.Y., Chen, H.S., Gerald, V.M., 2002. Development and implementation of wind-generated ocean surface wave modelsat NCEP. *Weath. Forecast.* 17 (2), 311–333.
- Torres, M.E., Colominas, M.A., Schlotthauer, G., Flandrin, P., 2011. A complete ensemble empirical mode decomposition with adaptive noise. In: *2011 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, pp. 4144–4147.
- Wang, A., Du, Y., Peng, S., Liu, K., Huang, R.X., 2018a. Deep water characteristics and circulation in the South China Sea. *Deep Sea Res. Part I* 134, 55–63.
- Wang, J., Liu, J., Wang, Y., Liao, Z., Sun, P., 2021. Spatiotemporal variations and extreme value analysis of significant wave height in the South China Sea based on 71-year long ERA5 wave reanalysis. *Appl. Ocean Res.* 113, 102750.
- Wang, Z., Li, S., Dong, S., Wu, K., Yu, H., Wang, L., Li, W., 2018b. Extreme wave climate variability in South China Sea. *Int. J. Appl. Earth Obs. Geoinf.* 73, 586–594.
- Wu, M., Stefanakos, C., Gao, Z., 2020. Multi-step-ahead forecasting of wave conditions based on a physics-based machine learning (PBML) model for marine operations. *J. Mar. Sci. Eng.* 8 (12), 992.
- Wu, Z., Huang, N.E., 2009. Ensemble empirical mode decomposition: a noise-assisted data analysis method. *Adv. Adapt. Data Anal.* 1 (01), 1–41.
- Wyatt, L., Thompson, S., Burton, R., 1999. Evaluation of high frequency radar wave measurement. *Coast. Eng.* 37 (3–4), 259–282.
- Yang, S., Deng, Z., Li, X., Zheng, C., Xi, L., Zhuang, J., Zhang, Z., Zhang, Z., 2021. A novel hybrid model based on STL decomposition and one-dimensional convolutional neural networks with positional encoding for significant wave height forecast. *Renew. Energy* 173, 531–543.
- Yeh, J.-R., Shieh, J.-S., Huang, N.E., 2010. Complementary ensemble empirical mode decomposition: a novel noise enhanced data analysis method. *Adv. Adapt. Data Anal.* 2 (02), 135–156.
- Ying, G., Zou, Y., Wan, L., Hu, Y., Feng, J., 2018. In: *Better guider predicts future better: Difference guided generative adversarial networks*, Asian Conference on Computer Vision. Springer, pp. 277–292.
- Yuan, C., Ma, X., Wang, H., Zhang, C., Li, X., 2023. COVID19-MLSF: a multi-task learning-based stock market forecasting framework during the COVID-19 pandemic. *Expert Syst. Appl.*, 119549.
- Zhang, C.A., Cho, S., Vasarhelyi, M., 2022. Explainable Artificial Intelligence (XAI) in auditing. *Int. J. Account. Inform. Syst.* 46, 100572.
- Zhou, F., Huang, Z., Zhang, C., 2022. Carbon price forecasting based on CEEMDAN and LSTM. *Appl. Energy* 311, 118601.